Tree Based Methods

Decision Tree:

version – ID3, C4.5, CART

Pruning → complexity কমাই

Visualization

* Explainable Rules
* Decision Tree example

Disadv- Decision Tree কে NP-complete problem বলে ।

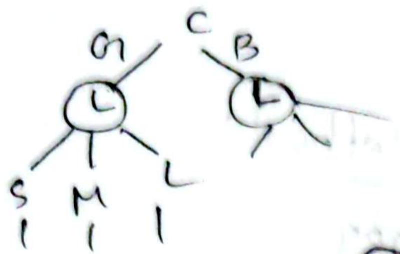
Tough for solving polynomial complexity is high.

O(N) N^2 N^3

$\boxed{\text{Decision Tree. feature importance}}$

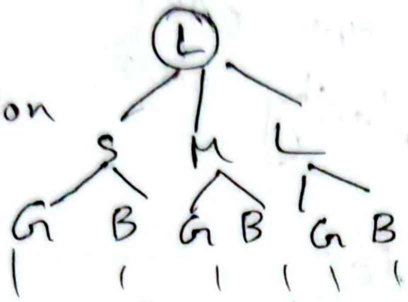
2 February 2026

* Handling Numeric Attributes
* Decision Tree



heuristic approach

what is good decision

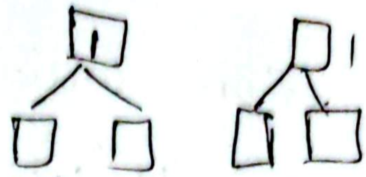


* Decision Tree for Regression and classification
  - classification and Regression Trees
    CART
    Use Gini measure of "impurity"
  - Iterative Dichotomiser 3
    Use information Gain, Greedy algorithm
  - C 4.5 by
    Pruning, attributes with different costs, missing values, continuous attributes

* Good split vs Bad split
  The case of classification
    Entropy, Information Gain, Gain Ratio, Gini Index

  Entropy – Measurement of disorder
    Minimize

Information Gain (IG) -



0.1 0.9

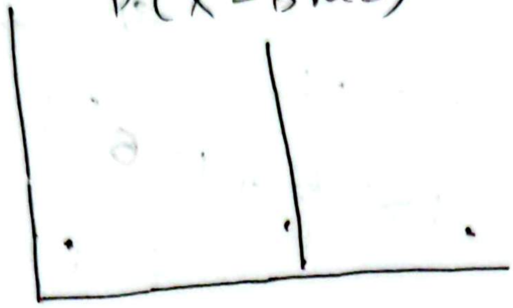
Previous / Initial Entropy

$IG = E_i - E_f = 1 - 0.1 = 0.9$

$1 - 0.9 = 0.1$

Gini Index (Gini Impurity) -

$P(X = Arg.) = 1$
$P(X = Braz.) = 0$



If we chose Argentina seper— 100%

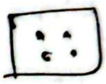
50% Arg 50% Bra

class probability

* Entropy

$$H(a) = -\sum_{i=1}^{K} P_i \log_2(P_i)$$

K = no. of class



$P(X = A) = 1$
$P(X = B) = 0$

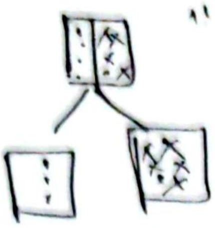
$H = -1 \log_2 1 - 0 \log_2 0$
$= 0$

In another case, 0.5 0.5

$H = -0.5 \log_2 0.5 - 0.5 \log_2 0.5$
$= 1$

* Information Gain

$$Gain(a, V) = H(a) - \sum_{i=1}^{|V|} \frac{N_i}{N_a} H(i)$$

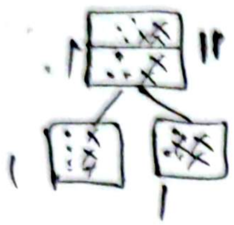


0 < entropy < 1

|V| → No. of split

$$-\frac{1}{7}\log_2\frac{1}{7} - \frac{6}{7}\log_2\frac{6}{7}$$

a = First stage entropy



$\cancel{1-(1+1) = -1}$

$$\cancel{\frac{3}{9}} \quad 1 - \left[1\times\frac{5}{11} + 1\times\frac{6}{11}\right]$$

i = New stage

N_i → split a count of a state

* Gain Ratio

Gain Ratio = Information Gain / Split Information

$$Split(a, V) = -\sum_{i-1}^{|V|} \frac{N_i}{N_a}\log_2\left(\frac{N_i}{N_a}\right)$$

Eg:- Credit card Number

ID	Location	CGPA	club
1	Haze Lane		
2	Outsid Hal		
3	H		
4	H		
5	O·H		

IG biases the decision tree against considering attributes with a large number of distinct values.

Entropy ?

* Gini Index

$$Gini(a) = \sum_{i=1}^{K} P_i (1-P_i)$$

$$Gain(a,v) = Gini(a) - \sum_{i=1}^{|v|} \frac{N_i}{N_a} Gini(i)$$

In exam Gini Index math probability more

* An example
* Outlook

8 Feb 2026

k = no. of class

* Gini Index

$$Gini(a) = \sum_{i=1}^{k} p_i(1-p_i)$$

Impt for exam

$$\Rightarrow Gini(a) = 1 - \sum_{i=1}^{k} (p_i)^2$$

* An example

Example link below the slide

[Sefiks: step by step cart decision tree example]

* Outlook
* Temperature
* Humidity
* Wind
* The first split
* The first split : outlook

Sun
hot
high
weak
No

100% accuracy

Depth completion
এর উপর কি রকম প্রভাব পড়তেছে
If less than
If more tha

Can ask in exam

* Recursive partitioning.
* outlook sunny & Temperature
* " sunny & wind
* The second split

* The final Tree
  Tempt - less important feature ⇒ Feature importance

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Math → from Gini Impurity, Entropy
Gini Impurity (CART)

* Decision Tree overfitting (overfitting কমানো কেমনে)
  → Pre-Pruning
    Max. no. of leaf nodes
    Max depth of the tree
    Min. no. of training instances at a leaf node.

  → Post-Pruning
    Use statistical Tests

* Detecting useless splits - Formula no need

VIMP * Decision Tree
  Pros & cons → need to learn

1 March 2026

Final – Decision Tree Math must come
Gini, Entropy

Random Forest

20,000 < 16,000 – Training
4,000 – Testing

20 Features

T_1 T_2 T_3 ... T_n

4k, 5F | 3k, 6F | 2k, 7F | 2F

classify করবে

0 1 0 1 1

voting এর মাধ্যমে final output দিবে।

choose the majority

n – estimators (কতগুলো tree নিবো)

Cross-validation

Hyperparameter optimize

K-Fold cross validation

20k dataset
↓
Subset এ ভাগ করবে

5-Fold

Hyper Parameter Optimization

20k
F-1 $\boxed{4k}$ → ~~Test~~/validation
F-2 $\boxed{4k}$ → Training
F-3 $\boxed{4k}$ – Tr
F-4 $\boxed{4k}$ – Tr
F-5 $\boxed{4k}$ – Tr

Average then result

Eg:- n_e = 100

Techniques:-

- Grid Search CV / Exhaustive search
- Randomized Search CV – Fast
- Bayesian Optimization (Very Fast)

Fast

RF

n_estimators = [50, 80, 100]
Depth = [20, 30, 50]
— gini, entropy

Grid search CV / Exhaustive search

n_e → Depth ↓	50	80	100
20	50,20	20,80	20,100
30		80,30	
50			

Feature selection
Recursive Dimension

• best_params –
80,30

SVM - Details, Math, Regression model - F1 score,
Classification precision, Recall
R-square value R-square Curve
Difference

Error Find → MAE, MSE, RMSE, when and how we use ?

2 March 2026

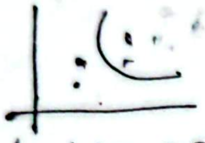
Support Vector Machines (SVM)

- The idea of SVM
- kernel Function

* Idea from - Vladimir ~~ladimik~~ Vapnik

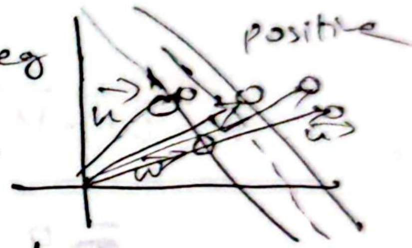
→ watch the video

* Decision Boundaries



Neural

$y = mx + c$



* Idea from -

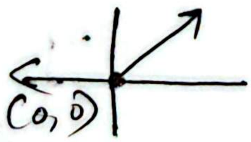
- We need a decision rule
- We need to learn the parameters of that street.

* Decision Rule

physics mathem

দিক
মান

cartesian system/plane starting from (0,0)



Computer-science vector

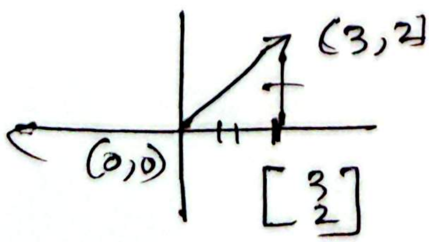
[2, 3, 10]

[1, 4, 1200, 2]

individual feature

1-D array
2-D matric R, C
3-D Frame, row, c

329 Linear Algebra playlist



$\begin{bmatrix} 3 \\ 2 \end{bmatrix}$

w → middle of street & perpendicular

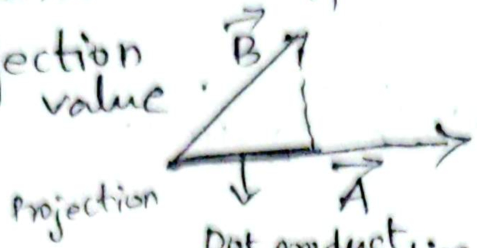
u → unknown instance

vector এর projection

– Dot product

$$\begin{bmatrix}3\\2\end{bmatrix} \begin{bmatrix}2\\1\end{bmatrix}$$

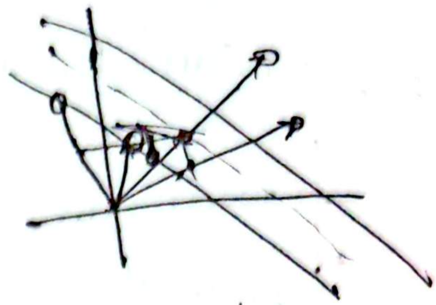
W·U = projection value



$$(3\hat{i}+2\hat{j})\cdot(2\hat{i}+\hat{j})$$

$6 + 1 = 7$

Dot product এর projection এর value দেয়।



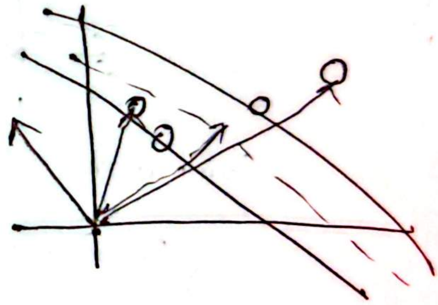
Projection Right – one class
Projection left → another class
Point

$w\cdot v >= c \quad \vec{w}\cdot\vec{u} \geq c$

c k v → constant

If $w\cdot u + b > 0$
label = +1
else
Label = −1

$y = mx + c$



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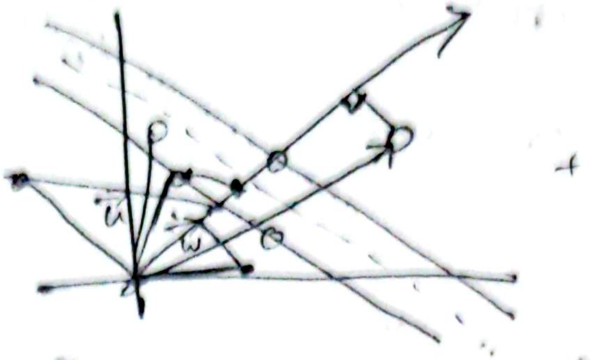
* Decision rule



$w \cdot v >= c$

$\vec{w} \cdot \vec{u} \geq c$

If $w \cdot u + b >= 0$

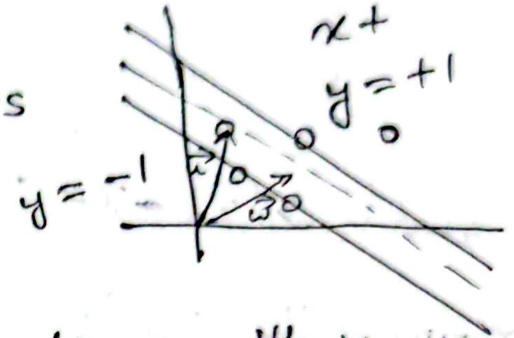
label = +1

else

label = −1

$c = -b$

* Support Vector Machines



$\vec{w} \cdot \vec{x}_+ + b \geq 1$

$\vec{w} \cdot \vec{x}_- + b \leq -1$

Let's multiply both the equations with y

$\vec{w} . \vec{x}_+ + b \geq 1$

$\vec{w} . \vec{x}_- + b \leq -1$

$\Rightarrow$

$y_i(\vec{w} \cdot \vec{x}_i + b) \geq 1$

$y_i(\vec{w} \cdot \vec{x}_i + b) \geq 1$

$\Downarrow$

$$\boxed{y_i(\vec{w} \cdot \vec{x}_i + b) - 1 \geq 0}$$

$y = +1$

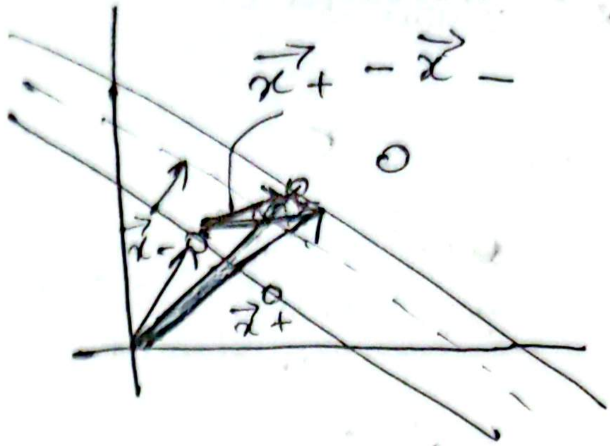
$y = -1$

~~test~~ corresponding label

unit vector $= \dfrac{\vec{A}}{|\vec{A}|}$

কোনো একটা vector এর direction

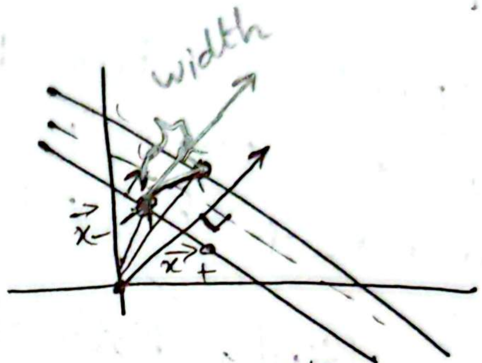
width বের করা



$\vec{c} = \vec{a} + \vec{b}$

$\vec{b} = \vec{c} - \vec{a}$

$$\text{width} = (\vec{x}_+ - \vec{x}_-) \cdot \frac{\vec{w}}{|\vec{w}|}$$



Maximize the width

$\downarrow$

$y_i(\vec{w} \cdot \vec{x}_i + b) = 1$

If $y_i = +1$ $\quad w \cdot x_+ + b = 1$

or, $w \cdot x_+ = 1 - b$

If $y_i = -1$

$-\vec{w} \cdot \vec{x}_- - b = 1$

$\Rightarrow \vec{w} \cdot \vec{x}_- = 1 - b$

$$\text{width} = (\vec{x}_+ - \vec{x}_-) \cdot \frac{\vec{w}}{||\vec{w}||}$$

$\downarrow$

$$\text{width} \left(\vec{x}_+ \cdot \vec{w} - \vec{x}_- \cdot \vec{w}\right) \cdot \frac{1}{||\vec{w}||}$$

$\downarrow \qquad\qquad \downarrow$

$1-b, \qquad 1+b$

$\downarrow$

$$\text{width} = \frac{2}{||\vec{w}||}$$

$\vec{w} \rightarrow$ width vector

We want to maximize this width

maximize $\dfrac{2}{\|\vec{w}\|}$

$\downarrow$

maximize $\dfrac{1}{\|\vec{w}\|}$

$\downarrow$

minimize $\|\vec{w}\|$

minimize $\frac{1}{2}\|\vec{w}\|^2$

condition

$y_i(\vec{w}.\vec{x_i}$

* Constrained optimization

How we do the optimization

optimize $f(x)$ with constraints $c(x)$, $d(x)$, ...

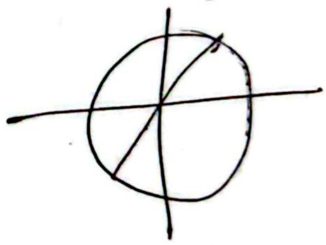
Suppose we wish to maximize the $f(x,y) = x+y$ subject to the constraint $x^2+y^2=1$

100

$x_{10} + y_{20} = 100$

maximize

$x^2+y^2=r^2$



constraint $g(x)$

$$\frac{x^2+y^2=r^2}{g(x)}$$

100

$x_{10} + y_{20} = f(x)$

$g(x) = 100$

width we can maximize constrain operation

* Lagrange Multipliers

$$L = f(x) + \lambda g(x)$$

$f(x)$ = width

$g(x)$ =

$$L = \frac{1}{2}\|\vec{w}\|^2 - \sum \alpha_i \left[y_i (\vec{w} \cdot \vec{x_i} + b) - 1 \right]$$

(constant value)

3 unknown value : w, α, b

$\boxed{\text{Till Pg 19}}$

Final এ আসতে পারে explanation ~~and derivati~~

Pg 5, 6, 7, 11, 12, 13, 14, 15, 16, 17, 18, 19,

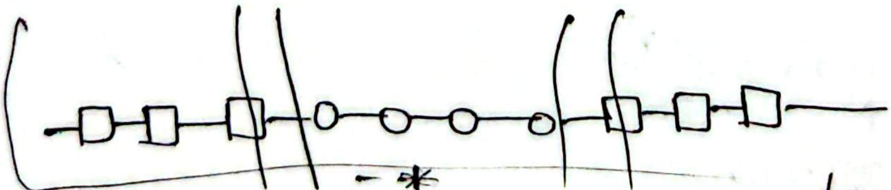
$\boxed{\text{watch the video Pg 1 link}}$ give.

CT-2 Decision Tree (Theory, math) Sunday (5 April 2026)

$\boxed{\text{29 March 2026}}$

* Support Vector Machines. — The kernel trick

1D — x এর value



যে dimensional data আছে separate করতে পারি না High dimensional এ নিয়ে যায়

High dimens - Support vector plane

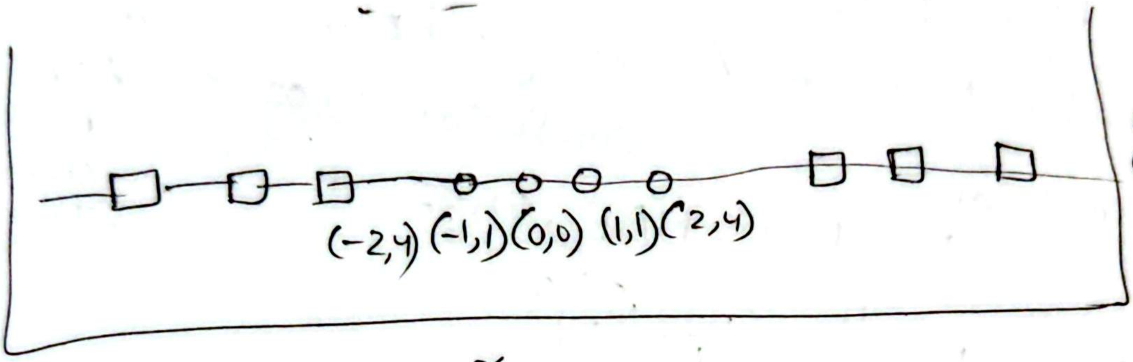
low dimension কে high dimensional এ নিয়ে যায় so that we can separate.

quadratic function

$$f(x) = ax^2 + bx + c$$

-2 4

$$y = x^2$$



* The kernel trick (mod 2)

Separate support vector

even - 0
odd - 1

* $y = x \% 2$

* The kernel trick (polinomial)

plane এ চলে যাবে

* The kernel trick

Linear
polynomial
Radial basis
sigmoid

$\phi(x)$ We donot need

$y = x^2$

$\phi(x)$

kernel $K(x_i, x_j) = \phi(x_i) \cdot \phi(x_j)$

$\vec{w} \cdot \vec{u}$

x এর value

Q1) কোন kernel টা কোথায় use করবো সুবিধা হবে for SVM?

Search in net

Hyperplane

Maximum Marginal Hyperplane

* Evaluation Metrics with formula

Classification — Accuracy, R^2, Precision, recall, log loss, confusion matrix, AUC and ROC curve, F1-Score

Regression

↳ MSE, MAE, RMSE

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2$$

$$MAE = \frac{\sum_{i=1}^{n} |y_i - x_i|}{n}$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2}$$

* Binary classification — Confusion matrix

		Actual N	Actual P
Predicted	N	TN	FN
	P	FP	TP

Formula.

Q) When to use which Metric (Binary classification)

6 April 2026

Regression & Classification

* Regression

data as x and labels as y.
$x \downarrow$ independent, $y \downarrow$ dependent

$x \; \{x_1, x_2, x_3, \dots x_n\}$ features $\longrightarrow$ $\boxed{\text{Regression model}}$ $\longrightarrow y$

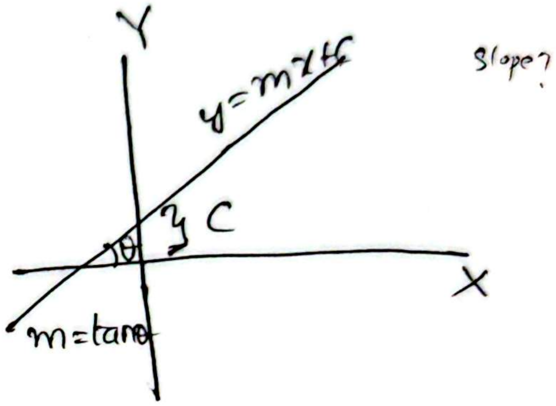
$y = f(x)$

$f(x) = ax + b$ ($x \to x^2$)
$f(x) = ax^2 + bx + c$
$f(x) = ax^3 + bx^2 + cx + d$

$f(x) = ax + b$ {straight line regression}

$\underline{y = ax + b}$
$y = mx + c$
$m =$ slope

$X \rightarrow$ feature विशेषता
$Y \rightarrow$ label

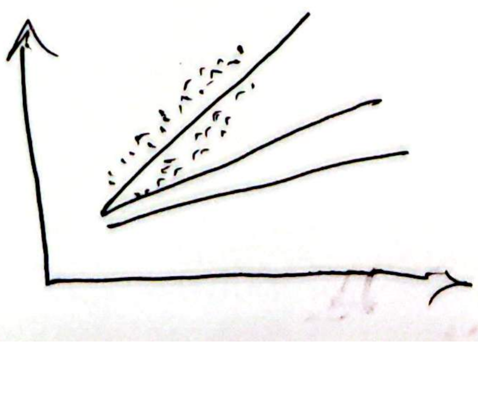


* Linear Regression

length $\longrightarrow$	Price
20	150
40	150
10	70
30	?

$y = w_0 + w_1 x_1$

* ~~Linear~~ $y = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3$



* Simple Linear Regression

• fit করা

w_0, w_1 বের করা

www.shiksha.com

$$m = \frac{\sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^{n} (x_i - \bar{x})^2} \quad \text{— (i)}$$

Testing dataset

x	y
x_1	y_1
x_2	y_2
x_3	y_3
x_4	y_4

$y = mx + c$

* and

$$c = \bar{y} - m\bar{x} \quad \text{— (ii)}$$

$\bar{y}$ = mean of y

$\bar{x}$ = mean of x

y_i - mean

Follow this approach but it's not there in slide

Memory where millions of data.

Mean Square

CE3

* Explanation of Error

$y = w_0 + w_1 x$

Actual ← y | $\hat{y}$ → predicted

Gradient descent

* Error in prediction

Squared error,

$$e = \frac{1}{2}\sum_{i=1}^{m} \left(\hat{y}(i) - y(i)\right)^2$$

Generates loss function, cost function

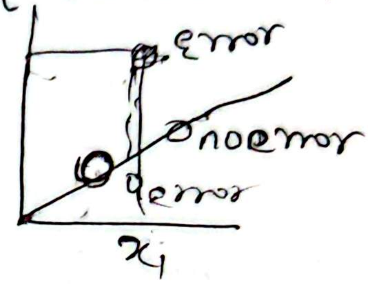
optimize loss function

single data নিয়ে কাজ

↓ পুরো dataset নিয়ে কাজ

Imp Exam

Q) Diff. b/w loss function & Cost function.



$i \to x_1, x_2$

Predicted, $\hat{y} = w_0 + w_1 x$

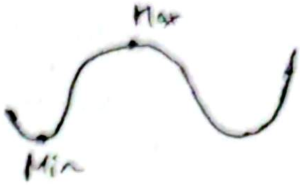
Actual $= y$

12 April 2026 — Linear Regression — CT-3

* Finding w_0, w_1

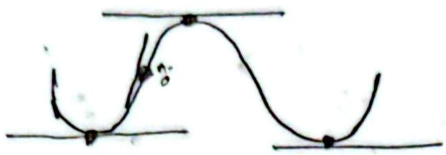
Minimize error

gradient descent algorithm

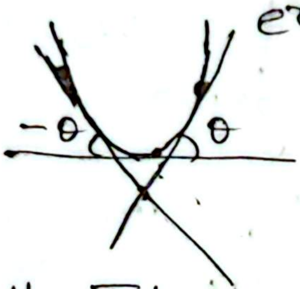


Find ~~ ~~ slope for min point

slope এর value 0



error curve



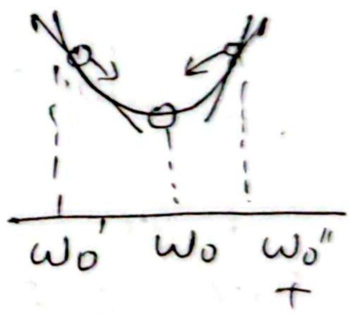
* ~~Intutt~~ Intution for Gradient Descent

$y = w_0 + w_1 x$

depend on slope

slope − বাড়াবো

slope + কমাবো



function slope $= \pm \dfrac{de}{dw_0}$ Sometimes −ve and sometimes +ve

$w_0(\text{new value}) = w_0(\text{old value}) - \alpha \dfrac{de}{dw_0}$

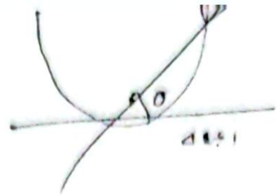
slope +ve

$w_1(\text{new value}) = w_1(\text{old value}) - \alpha \dfrac{de}{dw_1}$

* E

$\uparrow F \propto a \uparrow$

$\uparrow F = kma \uparrow$



Learning rate = 0.001
value vary
large learning rate: overshooting

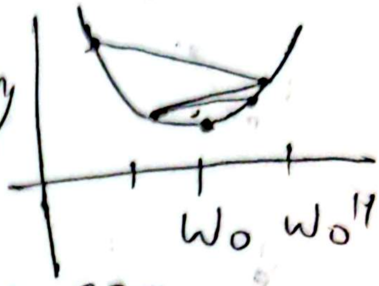
value of α or learning rate

Adam
RMSprop

* Learning Rate

Exam Q

Large learning rate: overshooting



Small learning rate:
Many iterations until convergence and trapping in local minima.

* Finding slopes

$$\frac{d}{dx}(2x^2) = 2 \cdot 2x$$

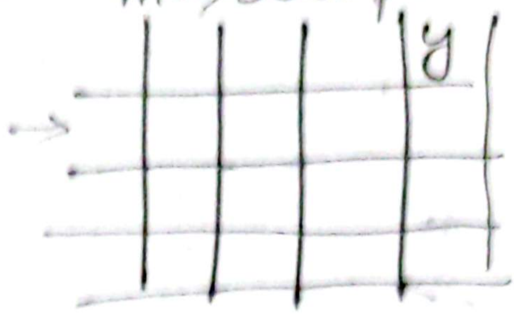
$$\text{slope} = \frac{\delta e}{\delta w_j} = \sum_{i=1}^{m} \left(\hat{y}(i) - y(i)\right) \cdot x_j(i)$$

$$e = \frac{1}{2} \sum_{i=1}^{m} \left((w_0 + w_1 x_1(i)) - y(i)\right)^2$$

* VIII Gradient Descent Algorithm

$n \rightarrow$ features

$m \rightarrow$ data points



$$y_i = w_0 + w_1 x_1(i) + w_2 x_2(i) + \dots + w_n x_n(i)$$

$$slope_j = slope_j + \underbrace{e}_{\text{error}} \times \underbrace{x_j(i)}_{\text{feature}}$$

slope update

$$w_j = w_j - \alpha \times slope_j$$

* Using Linear algebra

$$\begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1n} \\ 1 & x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{m1} & x_{m2} & & x_{mn} \end{bmatrix} \cdot \underbrace{\begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_n \end{bmatrix}}_{W}$$

$X \xrightarrow{n}$ = $m \times n$

Y = $m \times 1$

$X_E = m \times (n+1)$

Input X_E $\rightarrow$ Input Matrix $m \times (n+1)$

X_E = X extended

W $\rightarrow$ weighted Matrix $(n+1) \times 1$

$$y = w_0 + w_1 x_1 + w_2 x_2 - \dots -$$

$$\underset{\text{Predicted output}}{\hat{Y}} = \underbrace{\underset{m \times (n+1)}{X_E} \cdot \underset{(n+1) \times 1}{W}}_{m \times 1} \qquad \underset{(m \times 1)}{E} = \hat{Y} - Y$$

$$\begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1n} \\ 1 & x_{21} & & & \\ \vdots & \vdots & & & \\ 1 & x & & & \end{bmatrix} \cdot \begin{bmatrix} w_0 \\ w_1 \\ w \\ \vdots \\ w_n \end{bmatrix} = \begin{bmatrix} \\ \\ \\ \end{bmatrix} - \begin{bmatrix} \\ \\ \\ \end{bmatrix}$$

* $X \rightarrow (m \times n)$ $\quad X_E \rightarrow$ transpose

$X_E \rightarrow m \times (n+1)$

$W \rightarrow (n+1) \times 1$

$\hat{Y} \rightarrow (m \times 1)$ $\qquad X_E^T \rightarrow (n+1) \times m$

$E \rightarrow (m \times 1)$ $\qquad W \cdot - \begin{bmatrix} s_0 \\ s_1 \\ \vdots \\ s_n \end{bmatrix}$ slope

$S = X_E^T \cdot E$

$S \rightarrow (n+1) \times 1$

$$\boxed{\begin{aligned} \hat{Y} &= X_E \cdot W \\ E &= \hat{Y} - Y \\ S &= X_E^T \cdot E \end{aligned}}$$

15 * Using linear algebra

$$\underset{(n+1)\times 1}{W} = W - \alpha \times \underset{(n+1)\times 1}{S}$$

16 * Revised Gradient Descent

17 * More linear Algebra (কোডে নিয়ে)

22 * Logistic Regression

For classification we use Logistic Regression

21 * Experiments

$\hat{y} = mx + c$ $\qquad$ luxury $- 0$

$\hat{y} = w_0 + w_1 x$ $\qquad$ Economical $- 1$

$(-\infty, \infty)$ $\qquad$ o/p either 0 or 1

26 * step function

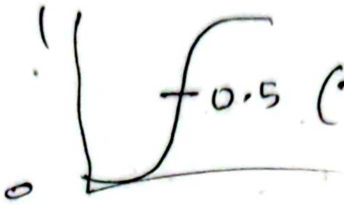
27
* Sigmoid Function

b/w 0 & 1 ↳ infinite পর্যন্ত যাবে

$$\sigma(\vec{x}) = \frac{1}{1+\exp(-\vec{x})}$$

$$\sigma(\hat{y}) = \frac{1}{1+e^{-\hat{y}}} \quad \text{near to } \infty \qquad \hat{y} = w_0 + w_1 x$$

$$e^{-100} = \frac{1}{e^{100}} \rightarrow 0 \text{ এর কাছাকাছি}$$



28
* A new loss function - Cross-entropy

$$e = \sum_{i=1}^{m} \left(-y(i)\log(\hat{y}(i)) - (1-y)\log(1-\hat{y})\right)$$

29-30
* Cross Entropy Loss
* 31 Comments on gradient descent

13 April 2026
Monday

Ensemble Methods

* How to select a movie?
* Approaches
  - voting
    - Majority
    - Average
    - Weighted
  - Bagging
  - Stacking
  - Cascading
  - Boosting
* Basic Assumption
  - Models should be different
  - Data should be different
* Majority voting
  - hard voting
* Hard voting vs soft voting

  যার % বেশি that class

* Base classifiers in Ensemble

Ensemble of classifiers
- → Ensemble of different classifiers
- → Ensemble of same classifiers

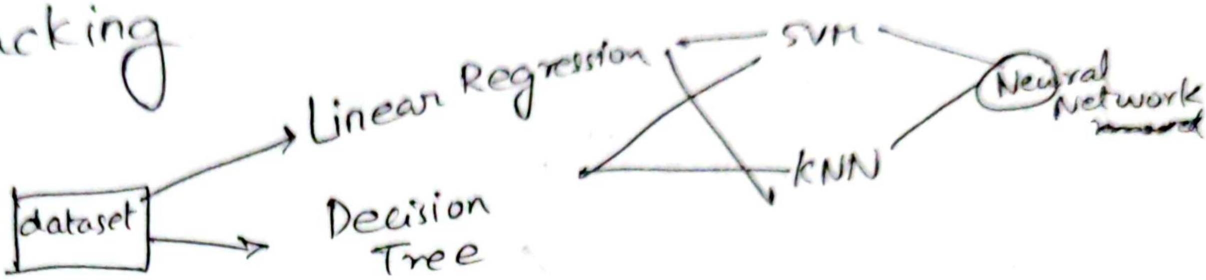
* Bootstrap Sampling
  Sampling with replacement

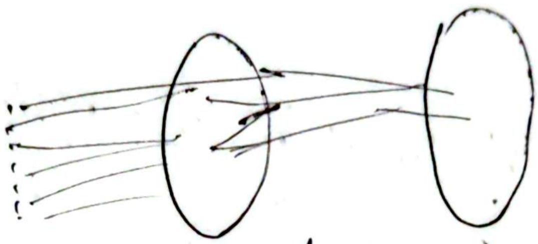
একবারে set এর data কে নিয়ে dataset এ রেখে

* একটি dataset থেকে multiple dataset হয়।

* Bagging == Bootstrap Aggregating

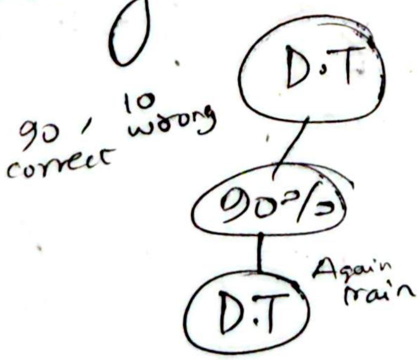
* Bagging
* Random Forest.
* Stacking





level 2 training data; M predictions are now feature

* Cascading



we can use Cascading in RNN.

* Boosting
  - Adaptive Boosting == AdaBoost

* A single classifier
  Error = e = percentage of mistakenly classified samples [0,1]

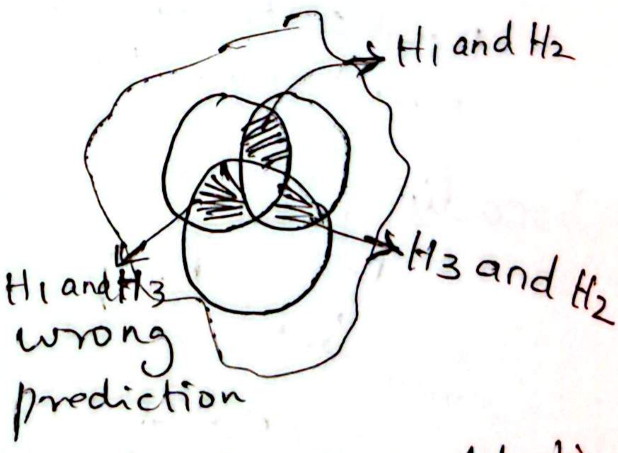
  $h(x)$ — hypothesis Algorithm

  1 == all wrong
  0 == all correct
  0.5 == random classifier

* A Weak classifier
  If error < 0.5 it is a weak classifier
  strong classifier → is more less than 0.5
  60-70

* Ensemble
  $H(x) = \text{sign}\,[\underbrace{h^1(x)}_{\text{wrong}} + \underbrace{h^2(x)}_{\text{wrong}} + \underbrace{h^3(x)}_{\text{wrong}}]$

overlapping problem
the ensemble performs worse than the individuals



মাঝখানের overlapping টা reduce হয় → Adaptive Boosting

Focus == weights

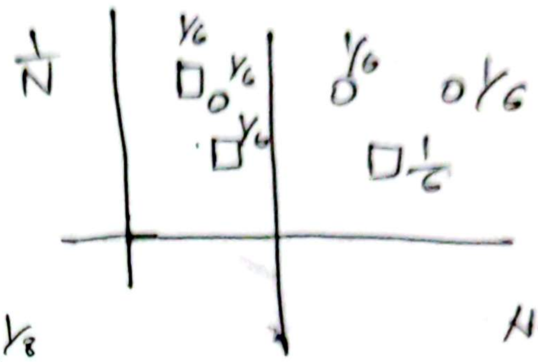
* Error Calculations

weight

$$e = \frac{1}{N} \sum_{wrong} 1 = \sum_{wrong} \frac{1}{N}$$

(Data'র ~~ওজন~~ weight) $w_i = \frac{1}{N}$

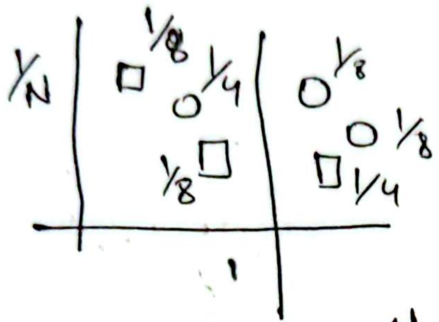
* Example



distribution

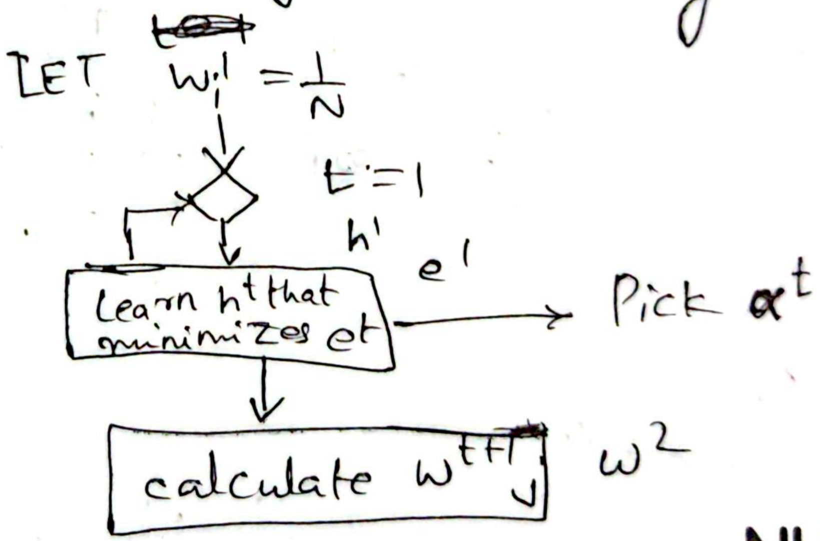
$$\sum_{all} w_i = 1$$

মোট summation 1.



1/4 right predict as weight more

* Iterative Algorithm: Boosting



zwei?

alpha weight

* Iteratively build the ensem...

$\left(\begin{smallmatrix} - & + \\ + & - \end{smallmatrix}\right)$ label

* Updating Weights

$$W_i^{t+1} = \frac{W_i^t * e^{-\alpha^t \boxed{h^t(x)} \boxed{y(x)}}}{Z}$$

$h^t(x) \downarrow$ prediction

$Z \searrow$ Some normalizing factor

$t+1$ = time का update करते हैं

i = rows

$\alpha \rightarrow$ Model का weight

+ +

− −

inverse $\frac{1}{e}$

* Min Error Bound

$$\alpha^t = \frac{1}{2} \ln \frac{1-e^t}{e^t}$$

$1-e^t \rightarrow$ right

$e^t \rightarrow$ wrong

e^t = error at t, iteration

$\log_e$

ln 0.001

−ve आएगा

>0 +ve

<0 −ve

w − punishment

α − reward

− correct

+ wrong

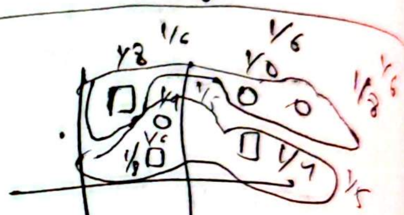
* Weight updates

* Without Z

$$\sum_{\text{Correct}} W_i^{t+1} = \sqrt{e^t(1-e^t)}$$

$e^t \downarrow$ error

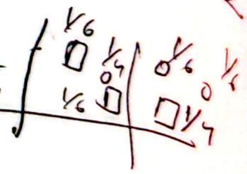
$(1-e^t) \downarrow$ right



$\frac{1}{8} : \frac{1}{8} : \frac{1}{8}$ $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$

1 : 1 : 1

$\frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$



$\frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6}$

$\frac{1}{8}$

* Sum Again

$$\sum_{\text{correct}} = \frac{1}{2}$$

$$\sum_{\text{wrong}} = \frac{1}{2}$$

$\frac{1}{4} : \frac{1}{8} : \frac{1}{4}$ [8 मिलेगा x]

2 : 1 : 2 $\frac{2}{5} \times \frac{1}{2} = \frac{1}{5}$

19 April 2026

* Using linear algebra

$$\hat{Y} = X_E \times W$$

$$E = \hat{Y} - Y$$

$$S = X_E^T \times E$$

$$W = W - \alpha \times S$$

* Problem statement

w_1 এর update value

x_1

w_0 w_1 w_2 w_3

$$z = \frac{1}{1-e^{-\hat{y}}}$$

CT-3 — step function, sigmoid function (continuous), Crossentropy (softmax function)

~~soft~~

— why step function is better than

— dataset weight update করা (math) (Ensemble)

* ML Taxonomy

- supervised
  - Regression
  - classification
- unsupervised
  - clustering
  - Association
- Semi-supervised = both labeled and unlabeled data.
  brain, cancer
- Reinforcement learning

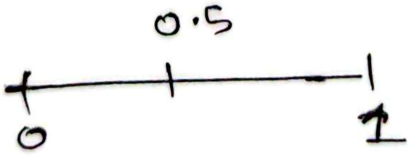
* Imbalanced dataset
  - Random over/under sampling
  - SMOTE
  - Cost sensitive classification

Credit card Fraud detection

confusion matrix

T.P	F.P
F.N	T.N

* What can you do?
* Random over/under sampling
* Problems with these approaches.
* Four groups of Negative Examples
  - Noise examples
  - Borderline examples



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* Tomek links [1]

যেটা কাছে
ঐটা tomek links

\+ + +
\+ +
+ + → distance is called Tomek links
\- -
\- -
\-

Majority class remove
balancing dataset

* Condensed Nearest Neighbor Rule (CNN rule)

Noise remove করে
1-NN rule

E — subset

$E' \subseteq E$

Positive – Minority class
Negative – Majority "

Tomek link use করে majority class বাদ দিবো

* Neighborhood cleaning Rule

3-NN

M (M) M
M

2 টা minority
2 টা Majority

Then remove the majority if Minority is more.

3-NN

M M
(m)
M

3 majority
1 Minority

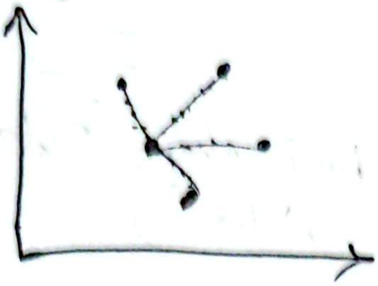
remove the majority

* SMOTE

Minority data class কে বানানো

SMOTE (Synthetic Minority Oversampling Technique)

Minority class বেড়ে যাবে।



* Main steps

(2,) —— • —— (4,5)

$2 + (4-2)\ 0.5$

$2 + 0.4(4-2)$

y coordinate

$1 + 0.4(5-1)$

$(2 + 0.4(4-2),\ 1 + 0.4(5-1))$

ϵ

near distance value

-2

then add

$(-\epsilon, \epsilon)$

Pure mathematics

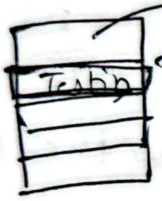
Date: 26 / 04 / 2026

Multi-view learning

Why the accuracy is ~~bett~~ better (99.9%)
oversampling, undersampling, Imbalanced dataset
CNN rule,

SMOTE
↓
Main steps

* Danger of information injection and over fitting
synthetic data এর আগে testing করবো
k-fold cross validation.
5 Fold
4 - Training
1 Testing.



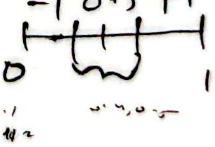
Testing
} ৪ টা synthetic data generate করে balance ~~করে~~ করার চেষ্টা)

* What can you do?
— cost sensitive classification

T.P	F.N
F.P	T.N

↳ Based on these cost implement

For classification
-1 0.5 +1



suppose logistic regression
0.5 ~~থেকে~~

Ornopain™

SMOTE

* classify as positive if : probability of positive

যেগুলো true না
ঐইগুলো true বলিতেছে

F.N

0	2
3	0

F.P
r

0.5 0.7

P

$$\boxed{\text{probability of positive} > \frac{r}{r+1}}$$

FP বেশি

$$\frac{3}{3+2} = \frac{3}{5}$$

0 0.5 1

0.34

Actually −ve
but model say
+ve

FN বেশি −

F.N

	3
2	

F.P

$$\frac{2}{3+2} = \frac{2}{5} < 0.5$$

* Performance metrics (see the formula)

Matthews correlation coefficient (MCC)

$$= \frac{TP \times TN - FP \times FN}{\sqrt{(\quad}}$$

Sa | Su | Mo | Tu | We | Th | Fr

Date:........./ ..../ ....

* Semi-Supervised Learning (SSL)

Disease classification

Both labeled and unlabeled

↓ small amounts

↓ large amounts

graph

o- unsupervised

L: a set of n labeled instances.

U: " " of m Unlabeled "

$n << m$

* SSL: self-learning

$y' = f(x)$ → for unlabeled

$x \in U$

$L = L \cup (x, y')$

$\frac{100}{f}$ $y' = f(x)$

10,000 → ① → x

101

100

$y'' f(x)$

* Self-learning improvements

* Multi-View learning

two independent sets of features or views.

Two classifiers

$x_1 \subset U \to h_1$

$x_2 \subset U \to h_2$

subset of unlabeled datasets.

Higher confident value add in $x_2 \subset U$, merge and then goes to h_2 same then which is better add in dataset

* Co-training algorithm চেহারা $\boxed{\text{GeeksforGeeks}}$

$\boxed{\text{Multi-view , Co-training}}$ কি কারণে হয়?

textual data imbalance থাকলে

backtranslation

* Assumptions
* How to combine the results?

* Tri-training (Zhou and Li)
Use 3 learners. If 2 agree, the data is used to teach the 3rd learner

$\begin{cases} h_1 \to 0.9 \\ h_2 \to 0.88 \end{cases}$

h_3

From this chpt 1 set Q will come for final exam,

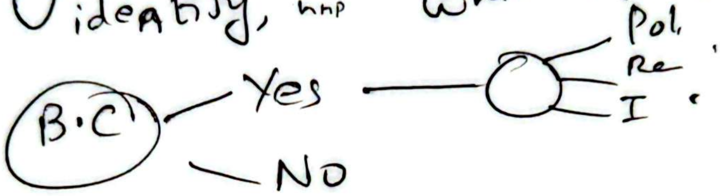
- Decision Tree (Math একটা থাকবে IG, Entropy, Gini Impurity)
- SVM (Basic কাজ, কি, no math), Diagram
- Till Lagrange Multiplier
  kernel trick polinomial, linear, sigmoid, Radial basis কি করে?

Sa | Su | Mo | Tu | We | Th | Fr

Date:........../ ..../ ....

— Linear Regression, Logistic regression
Math pdf link, step function use.
continuous, sigmoid use,
cross entropy (formula same) হয়ে যায়
linear algebra (formula)

— Ensembling
voting, cascading, Bootstrap sampling idea,
Bootstrap থেকে Bagging হিসাবে আসছে, Random Forest
Stacking vs Cascading (Diff.)
identify, Final predi hnp which type (political, religious)



In this case we use stacking

Boosting (math আসতে পারে)
- Adaptive Boosting
(pg21) Example (21, 22)
updating weights (24),
Min Error Bound (25) } → Formula

Math pdf sir will provide

Mathematical, Reasoning Types (more)

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